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RESEARCH ARTICLE

Effect of microplastics on *Yersinia ruckeri* infection in rainbow trout (*Oncorhynchus mykiss*)

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Abstract

Exposure to microorganisms such as *Yersinia ruckeri* can significantly affect bacterial infections in fish. Microplastics (MPs) may predispose fish to infection and act as carriers in pathogen transmission. Therefore, this study is designed to evaluate MPs' effect on damage caused by exposure to *Y. ruckeri* in rainbow trout. In this study, blood biochemical parameters and hepatic oxidative biomarkers as clinical signs were measured in the fish co-exposed to *Y. ruckeri* (5 and 10% the median lethal dose (LD_{50})) and MPs (500 and 1000 mg Kg^{-1}) for 30 days. There were no significant changes in the creatinine, triglyceride, cholesterol levels, and glutamic-pyruvic transaminase activity in the blood of fish infected with *Y. ruckeri*. In contrast, exposure to MPs had a significant effect on most clinical parameters. The total protein, albumin, globulin, total immunoglobulins, high-density lipoprotein, low-density lipoprotein, cholesterol levels, and γ-glutamyltransferase activity decreased, whereas glucose, triglyceride, and creatinine levels, and glutamic-oxaloacetic transaminase, glutamic-pyruvic transaminase, alkaline phosphatase, and lactate dehydrogenase activities increased in the plasma of fish after co-exposure to MPs and *Y. ruckeri*. Dietary MPs combined with a *Y. ruckeri* challenge decreased catalase and glutathione peroxidase activities, and total antioxidant levels. However, superoxide dismutase activity and malondialdehyde contents increased in the hepatocyte of fish co-exposed to MPs and *Y. ruckeri*. This study suggests that fish exposure to MPs and simultaneous challenge with *Y. ruckeri* could synergistically affect clinical parameters.

Keywords Yersinia ruckeri · Microplastic · Blood biochemical parameters · Oxidative stress · Rainbow trout

Introduction

Microplastics (MPs) are known as a newfound and unique pollutant in aquatic ecosystems (Kim et al. 2021). The universal expansion of the plastics industry and the increase of consumerism have drastically increased garbage and plastic waste in the environment (Iheanacho et al. 2020; Zhang et al. 2021). Destruction of plastic debris under the influence of UV, physicochemical changes in the environment, and physical processes have caused most plastic compounds to break down to MPs and enter aquatic ecosystems through surface runoff (Frias and Roisin Nash 2019). MPs may enter the fish body through the food chain or absorption by the gills. MPs accumulate in aquatic animals, get into cells, and travel through the body via the circulatory or lymphatic system (Wang et al. 2020; Kim et al. 2021). MPs often get into the digestive system and are transmitted to the liver or other vital organs (Jovanović 2017). Therefore, accumulation of MPs in vital tissues of fish can have toxic effects on growth and development, immune system functions, oxidative stress, metabolism and energy budgeting, and performance of various biomarkers (Qiao et al. 2019a, b; Bhagat et al. 2020).

Prolonged exposure to MPs can lead to an inflammatory response in the gut, leading to metabolic disorders and diseases caused by imbalance and damage to the microbial flora of fish intestines (Kang et al. 2021). Studies show that MPs can have detrimental effects on aquatic health (Brandts et al.

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2018; Ding et al. 2018). Exposure of aquatic organisms to MPs can lead to oxidative stress, suppression of the immune system, changes in blood biochemical parameters, histopathological damage, and gene expression changes (Chen et al. 2020; Pannetier et al. 2020; Ding et al. 2020; Banaee and Shakeri 2021).

Besides contaminating the aquatic ecosystems, MPs have a high potential for transporting environmental and biological pollutants. Various researchers have studied the MPs' synergistic effect on increasing the toxicity of heavy metals, pesticides, and aromatic petroleum hydrocarbons (Nematdoost Haghi and Banaee 2017; Banaee et al. 2019a; Kim et al. 2021). These studies show that MPs can have a significant effect on the bioavailability of environmental pollutants.

Although MPs' effect on the pathogenicity of pathogens has not been studied extensively, there is a hypothesis that exposure to MPs may suppress the immune system of fish and predispose them to pathogens.

Yersinia ruckeri is an opportunistic pathogenic gram-negative bacterium known to cause enteric redmouth disease (Gültepe 2020; Terzi et al. 2021). The spread of this bacterium among different commercial fish species can cause significant damage to the aquaculture industry (Raida and Buchmann 2008). *Y. ruckeri* has been isolated from various species of fish in many parts of the world. *Y. ruckeri* is transmitted to fish via water, food, sediments, and other aquatic animals. The Yersinia's infection is often transmitted through the fecal-oral route (Dezfuly et al. 2020; Terzi et al. 2021).

Y. ruckeri is an invasive microorganism that can induce infection via tissue destruction. The pathogen passes into the digestive system through contaminated feed and penetrates the intestine's epithelial cells. *Y. ruckeri*'s infection may lead to mucosal ulcers and necrotic lesions in the digestive system of fish. The enterotoxin synthesized by *Y. ruckeri* may play a significant role in causing disease (Dezfuly et al. 2020; Terzi et al. 2021). After challenging fish with bacteria, *Y. ruckeri* can enter fish blood through gill lamella. The infection affects the intestine, liver, spleen, heart, and brain within 30 min to 3 days (Kumar et al. 2015). Although the presence of *Y. ruckeri* from 1 h to 5 days after the challenge was confirmed by polymerase chain reaction (PCR) analysis, the bacterial infection in the blood was undetectable from day 9 to 30 (Altinok et al. 2001). Furthermore, the bacteria were no longer detectable in these tissues 21 days after the initial challenge (Kumar et al. 2015).

Since environmental conditions play a crucial role in the prevalence of opportunistic pathogens, the question is whether MPs can predispose fish to yersiniosis. Therefore, assessing the health of fish co-exposed to MPs and *Y. ruckeri* may help answer this question. Blood biochemical parameters are a good indicator for assessing the health of organs involved with pathogens and xenobiotics. Oxidative stress biomarkers also indicate the cellular antioxidant defense system's power against free radicals due to xenobiotics and biological toxins' metabolism. Thus, this study was designed to study fish health based on blood biochemical parameters and oxidative stress biomarkers. Rainbow trout is one of the commercial and farmed species in Iran that is very susceptible to Yersinia infection. In this study, the rainbow trout were used to investigate the clinical signs of *Y. ruckeri* pathogenesis when facing MPs.

Materials and methods

Chemical materials

High-density polyethylene (PE100) powder (< 0.02 mm) was purchased from Eshragh Trading Co., Iran. All diagnostic kits used to measure biochemical parameters and antioxidant enzymes were obtained from Pars Azmun Co. and BiorexFars Co., Iran.

Fish

Juvenile rainbow trout were purchased from a private farm and transferred to the Aquatic Animal Health Division (School of Veterinary Medicine, Shiraz University, Iran). The fish were adapted to laboratory conditions (temperature 15 ± 2 °C; photoperiod cycle 14 light/8 dark; dissolved oxygen 8.1 ± 0.7 mg L^{-1}; pH 7.3 ± 0.4; electrical conductivity 693.83 ± 76 µS cm^{-1}; salinity 0.3 ± 0.02 g L^{-1}) for 2 weeks. The School of Veterinary Medicine's ethical committee confirmed this study's proposal and all the experimental procedures.

The median lethal dose (LD_{50}) of Yersinia ruckeri for rainbow trout

The *Yersinia ruckeri* (PTCC No: 1888) was cultured in Trypticase Soy Agar with 5% defibrinated sheep blood at 25 °C for 48 h. Next, bacteria were washed twice using sterile phosphate-buffered saline (PBS). The bacterial concentration was adjusted to 10^{10} CFU ml^{-1} using the pour plate count method.

Rainbow trout were randomly distributed into eighteen tanks (9 trial groups in two replicates). Each tank contained fifteen fish. Eight serial dilutions of *Y. ruckeri* suspension (10^3 to 10^{10} cells ml^{-1}) were added to each tank (10 L) for 1 h. The fish in the control group were maintained in dechlorinated tap water. Clinical signs and mortality rate were recorded following the challenge with bacteria for 7 days. During the LD_{50} experiment, water was aerated and had the same conditions as the acclimation period. The dead fish were immediately removed from the tanks. Mortality due to bacterial infection was confirmed after sampling and bacterial culture in vitro. Seven days LD_{50} (i.e., lethal dose affecting 50 % of a population of animals) was determined by probit analysis.

The experimental challenge trials

Two hundred and seventy rainbow trout (15 ± 2.5 cm; 24.4 ± 6.3 g) were randomly distributed in twenty-seven fibreglass tanks (100 L) into nine separate experimental groups in triplicate. The 1st group or control group was maintained in tap water (without MPs and *Y. ruckeri*). The 2nd and 3rd groups were immersed in a suspension of *Y. ruckeri* at concentrations of 5% (2.5×10^5 CFU L^{-1}) and 10% (5×10^5 CFU L^{-1}) LD_{50} for 1 h, respectively. The 4th and 5th groups were exposed to 500 and 1000 mg Kg^{-1} of MPs, respectively. The concentration of MPs was determined based on a study (Zhu et al. 2020). The 6th group was treated with 5% LD_{50} of *Y. ruckeri* and exposed to 500 mg Kg^{-1} of MPs. The 7th group was challenged with 5% LD_{50} of *Y. ruckeri* and exposed to 1000 mg Kg^{-1} of MPs. The 8th group was treated with 10% LD_{50} of *Y. ruckeri* and exposed to 500 mg Kg^{-1} of MPs, and the 9th group was challenged with 10% LD_{50} of *Y. ruckeri* and exposed to 1000 mg Kg^{-1} of MPs. Fish were continuously exposed to MPs for 30 days. However, fish were rechallenged by a bath immersion of *Y. ruckeri* cells, 8, 15, and 22 days after clearance of the first challenge. The fish were rechallenged to ensure their exposure with *Y. ruckeri* throughout the experiment (Raida and Buchmann 2008; Raida and Buchmann 2009). The bacterial dose was considered based on the minimum LD_{50} (5.6×10^6 CFU L^{-1}). Before the rechallenge, *Y. ruckeri* load was assessed in blood to ensure bacterial clearance (Altinok et al. 2001). Every 24 h, water was renewed (100%), and fresh suspension of MPs was prepared and added to water to maintain the nominal dose. In the experimental periods, rainbow trout were fed twice a day with commercial feed. One day before sampling, fish feeding was stopped. At the end of the challenge period, the fish were anaesthetized using clove powder (150 mg L^{-1}). The blood was sampled from the stem vein using a 2.5-mL syringe impregnated with an anticoagulant. The blood samples were centrifuged (6000*g*, at 4° C for 15 min), and plasma fractions were collected and stored at − 25° C until the biochemical analyses.

After the sacrifice, the fish were dissected, and then the liver was quickly dissected and homogenized on ice in 100 mM cold potassium phosphate buffer (Sigma-Aldrich, Germany) pH 7.0, with 2 mM EDTA (Riedel-Haën, Germany). Tissue homogenates were centrifuged at 12,000*g*-force for 15 min at 4 °C, and the supernatants were collected and maintained at − 80 °C until further analysis.

Blood biochemical analysis

Glutamic-oxaloacetic transaminase (SGOT), glutamic-pyruvic transaminase (SGPT), alkaline phosphatase (ALP), γ-glutamyltransferase (GGT), and lactate dehydrogenase (LDH) activities, glucose, total cholesterol, triglyceride, high-density lipoprotein (HDL), low-density lipoprotein (LDL), creatinine, total protein, and albumin levels in plasma were evaluated based on the standard methods provided in the diagnostic kit protocol. The globulin levels were also computed based on the following formula:

$$\text{Globulins} = \text{Total protein–Albumin}$$

Total immunoglobulin (Ig) was estimated using polyethylene glycol solution according to the procedure described by Banaee et al. (Banaee et al. 2019b). All blood biochemical parameters were measured utilizing a UV-visible spectrophotometer (Unico, 2100) and standard biochemical reagents obtained from Pars-Azemun Co, Tehran, Iran.

Oxidative biomarker analysis

Superoxide dismutase (SOD) and glutathione peroxidase (GPx) activities in the hepatocytes were assessed using oxidative biomarker kits purchased from the Biorexfars Co., Iran. The total antioxidant capacity was estimated with the ferric reducing ability of plasma (FRAP) technique using TPTZ (2,4,6-Tris(2-pyridyl)-s-triazine) as a substrate (Benzie and Strain 1996). Per-oxidative damage in the hepatocytes was evaluated in malondialdehyde (MDA) production by the method presented by Placer and others (Placer et al. 1966), using thiobarbituric as substrate acid and monitored at 532 nm. Catalase (CAT) activity was measured spectrophotometrically in tissue homogenates by using a hydrogen peroxide solution (30 mM) as a substrate and monitored at 450 nm (Góth 1991).

Assessment of synergism and antagonism

In the present study, the following mathematical models were used to predict the interaction of MPs and *Y. ruckeri* on each other. According to Banaee et al. (2021), the synergism rate is calculated only when the predicted effect is higher than the observed one.

1. Predicted effect of the endpoints of rainbow trout exposed to either MPs or *Y. ruckeri*

$$\text{Predicted effect} = \frac{\text{Y.ruckeri}}{\text{Control}} \times \frac{\text{MPs}}{\text{Control}}$$

2. The observed effect of the endpoints of rainbow trout exposed to both *Y. ruckeri* and MPs

$$\text{Observed effect} = \frac{\text{Y.ruckeri combined with MPs}}{\text{Control}}$$

3. The synergism effect

$$\text{Synergy ratio} = \frac{\text{Predicted effect}}{\text{Observed effect}}$$

Data analysis

Statistical analyses of all data were carried out using Graph Pad Prism 8.0.2. The Shapiro-Wilk test calculated the normality of all data. Then, changes among different variables of rainbow trout exposed to *Y. ruckeri* combined with MPs were analyzed using a two-way analysis of variance (two-way ANOVA). A Tukey test was used to compare differences in each parameter between control and experimental groups. A significant difference was assessed at $P < 0.05$ levels, and results were shown as mean ± standard deviation.

Results and discussion

Clinical signs such as hemorrhage on the body, especially at the base of the fins and anus, bleeding and inflammation in the intestine's distal parts, petechiae bleeding around the jaw and mouth, bleeding or pale gills were observed in the experimentally infected rainbow trout. We isolated *Y. ruckeri* from the intestine, kidney, liver, and gill of all recently dead and survived fish. A cumulative mortality rate of 60%, 70%, 80%, and 100% was reached in rainbow trout immersed in *Y. ruckeri* suspension at doses of 10^7, 10^8, 10^9, and 10^{10} CFU L^{-1}, respectively, and LD_{50} value for *Y. ruckeri* was estimated to be 2.1×10^7 CFU L^{-1} (5.6×10^6–8×10^7 CFU L^{-1}). Kumar et al. (2015) showed that the lethal dose of *Y. ruckeri* depends on the expression of virulence genes. In this study, 5% (2.5×10^5 CFU L^{-1}) and 10% (5×10^5 CFU L^{-1}) LD_{50} doses were selected based on the lowest LD_{50} dose level (5.6×10^6 CFU L^{-1}).

Although none of the fish used in the experiment died during co-exposure to MPs and *Y. ruckeri*, clinical signs of yersiniosis were observed in fish. The pathogenicity of *Y. ruckeri* and fish mortality may be attributed to virulence factors. The most well-known virulence factors are lipopolysaccharides, hemolysin, Yrp1 protease, ruberbactin, and extracellular metalloproteinase (Portnyagina et al. 2021). Furthermore, *Y. ruckeri* can damage host cellular membranes through secreting nonspecific pore-forming proteins (porins) (Portnyagina et al. 2021).

Glucose is an important energy source in fish, especially in the brain, which gets most of its energy from carbohydrates. However, fish is known as a species with low glucose tolerance. Exposure of fish to MPs and *Y. ruckeri* led to elevated glucose levels (Fig. 1 and Table 1). Elevated glucose may be a physiological response to increased energy demand to combat biotoxins and MPs. Bao et al. (2020) showed that the liver's glucose metabolism gene altered after xenobiotic exposure. An increase in blood glucose may also increase the conversion rate of excess glucose to triglycerides and increase its level. An increase was observed in the blood glucose of tilapia *O. niloticus* (Hamed et al. 2019), and *C. carpio* (Banaee et al. 2019b), after exposure to MPs. Elevated glucose may be related to plasticiser's effect on insulin resistance and glucose metabolism (Carlsson et al. 2018). Damage to pancreatic beta cells can change blood glucose homeostasis (Ropero, et al., 2008). Plasticisers can interfere with the cellular signaling pathway in maintaining blood glucose homeostasis (Stojanoska et al. 2017; Han et al. 2019). The highest glucose levels were observed in the blood of fish exposed to 1000 mg Kg^{-1} MPs and challenged with 5% LD_{50} *Y. ruckeri*. This result suggests that MPs and *Y. ruckeri* exert a synergistic effect by increasing glucose levels.

Creatinine is the most critical metabolite produced by the muscles and excreted through the kidneys. The concentration of creatinine in the blood of fish exposed to 1000 mg Kg^{-1} MPs increased significantly compared to the control group. However, there was no significant change in creatinine level in fish fed with 500 mg Kg^{-1} MPs (Fig. 1 and Table 1). Exposure to MPs could impair the kidney and decrease its ability to secrete creatinine by reducing glomerular filtration. A study done by Hamed et al. (2019) showed that exposure of tilapia *O. niloticus* to MPs increased creatinine levels (Hamed et al. 2019). No significant differences were found in the creatinine levels in fish after being challenged with *Y. ruckeri* alone. However, creatinine concentration in fish co-exposed to both MPs and bacterial infections was significantly higher than the control group. This study showed that fish exposure to MPs challenged with *Y. ruckeri* could lead to a synergic effect on creatinine levels.

Results showed that bacterial infection of fish did not cause a significant change in triglyceride concentration. Fish exposure to MPs resulted in a significant increase in triglyceride concentration compared to the control group (Fig. 2 and Table 1). Simultaneous exposure of fish to MPs and challenge with *Y. ruckeri* significantly increased triglyceride concentrations. Bao et al. (2020) found that exposure of fish to xenobiotics could alter genes' expression in lipid metabolism and adipogenesis (Bao et al. 2020). Moreover, increased metabolic expenditure after *Y. ruckeri* challenge and MPs exposure to maintain homeostasis and detoxification can lead to changes in metabolite concentrations (Wang et al. 2021). The glycerol in triglycerides can be converted to glucose (Wen et al. 2021). Therefore, an increase in triglycerides

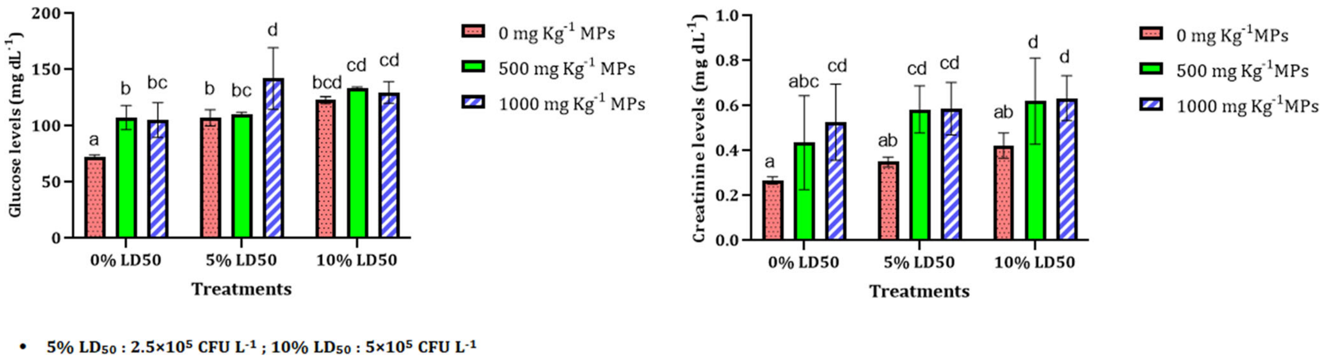


Fig. 1 Changes of glucose and creatinine levels in the blood of rainbow trout after exposure to MPs and *Y. ruckeri*. Results are illustrated as mean ± SD. Significant differences between groups were identified by alphabetical characters ($P < 0.05$). Different letters of the alphabet indicated the existence of significant differences between different groups

may indirectly increase the blood glucose of fish exposed to MPs and *Y. ruckeri*. This study revealed that fish exposure to MPs combined with *Y. ruckeri* could lead to a synergic effect on the triglyceride levels. The triglyceride levels were elevated in the serum of common carp, *C. carpio* and freshwater pond turtles, *E. orbicularis*, after treatment with MPs (Banaee et al. 2019a, 2021).

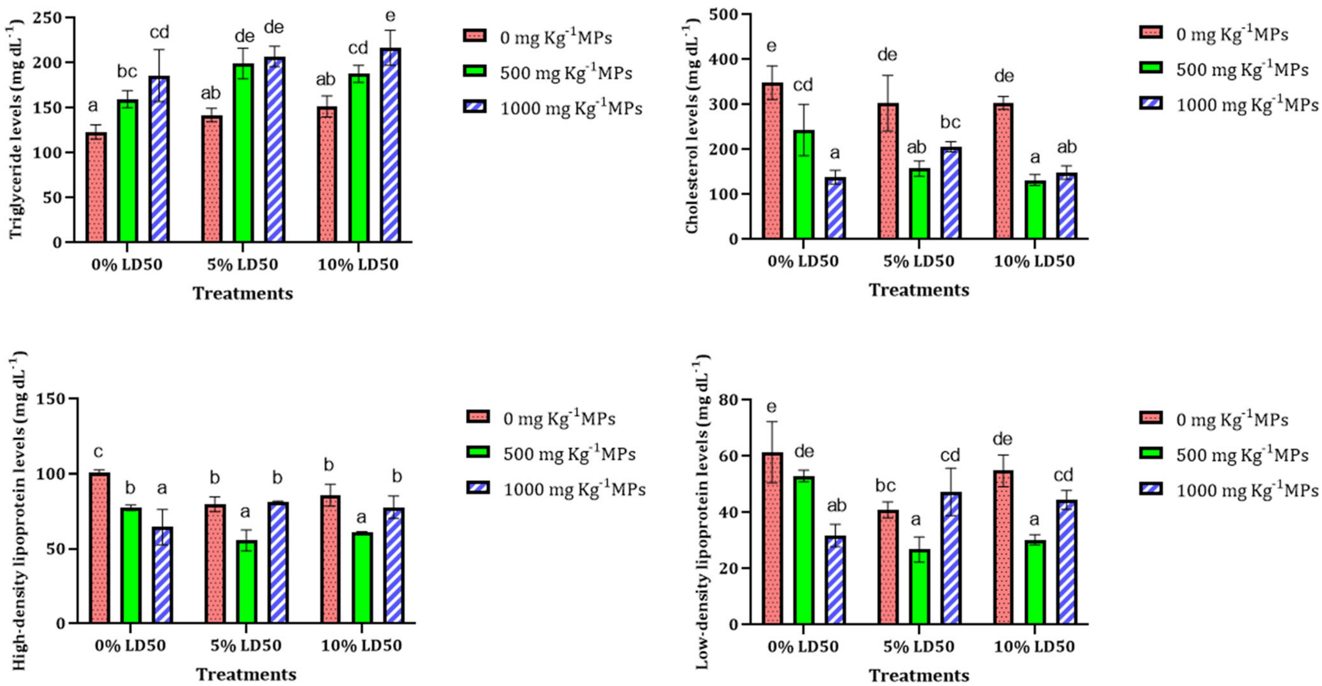
There were no significant changes in the cholesterol levels in the blood of fish infected with *Y. ruckeri*. Treatment of fish with MPs alone and MPs combined with *Y. ruckeri* resulted in a significant reduction in cholesterol concentration. This study showed that fish co-exposure to MPs and *Y. ruckeri* could lead to a suppressive effect on cholesterol levels (Fig. 2 and Table 1). Changes in the blood cholesterol concentration in

Table 1 Evaluation of synergism of MPs and *Y. ruckeri* concerning the blood biochemical parameters

Parameters	Treatment	PE	OE	SR	CE	Parameters	Treatment	PE	OE	SR	CE
Glucose	500 MPs+ 5% LD50	2.21	1.53	1.45	S	LDL	500 MPs+ 5% LD50	0.57	0.44	1.30	S
	1000 MPs+ 5% LD50	2.16	1.97	1.10	S		1000 MPs+ 5% LD50	0.34	0.77	0.44	A
	500 MPs+ 10% LD50	2.53	1.85	1.37	S		500 MPs+ 10% LD50	0.77	0.49	1.56	S
	1000 MPs+ 10% LD50	2.48	1.80	1.38	S		1000 MPs+ 10% LD50	0.46	0.72	0.64	A
Creatinine	500 MPs+ 5% LD50	2.12	2.18	0.97	A	Total protein	500 MPs+ 5% LD50	0.64	0.66	0.97	A
	1000 MPs+ 5% LD50	2.56	2.19	1.17	S		1000 MPs+ 5% LD50	0.55	0.68	0.80	A
	500 MPs+ 10% LD50	2.56	2.31	1.11	S		500 MPs+ 10% LD50	0.61	0.66	0.92	A
	1000 MPs+ 10% LD50	3.10	2.36	1.31	S		1000 MPs+ 10% LD50	0.52	0.72	0.72	A
Triglyceride	500 MPs+ 5% LD50	1.49	1.62	0.92	A	Albumin	500 MPs+ 5% LD50	0.89	0.78	1.14	S
	1000 MPs+ 5% LD50	1.73	1.68	1.03	S		1000 MPs+ 5% LD50	0.79	0.85	0.93	A
	500 MPs+ 10% LD50	1.59	1.52	1.05	S		500 MPs+ 10% LD50	0.96	0.81	1.19	S
	1000 MPs+ 10% LD50	1.85	1.76	1.05	S		1000 MPs+ 10% LD50	0.85	0.93	0.92	A
Cholesterol	500 MPs+ 5% LD50	0.61	0.45	1.34	S	Globulins	500 MPs+ 5% LD50	0.49	0.57	0.86	A
	1000 MPs+ 5% LD50	0.35	0.59	0.58	A		1000 MPs+ 5% LD50	0.40	0.57	0.70	A
	500 MPs+ 10% LD50	0.61	0.38	1.60	S		500 MPs+ 10% LD50	0.41	0.56	0.74	A
	1000 MPs+ 10% LD50	0.35	0.43	0.81	A		1000 MPs+ 10% LD50	0.34	0.57	0.59	A
HDL	500 MPs+ 5% LD50	0.61	0.55	1.10	S	Total immunoglobulins	500 MPs+ 5% LD50	0.56	0.38	1.46	S
	1000 MPs+ 5% LD50	0.51	0.80	0.63	A		1000 MPs+ 5% LD50	0.45	0.41	1.09	S
	500 MPs+ 10% LD50	0.65	0.60	1.08	S		500 MPs+ 10% LD50	0.56	0.32	1.73	S
	1000 MPs+ 10% LD50	0.54	0.77	0.70	A		1000 MPs+ 10% LD50	0.45	0.36	1.23	S
LDL	500 MPs+ 5% LD50	0.57	0.44	1.30	S						
	1000 MPs+ 5% LD50	0.34	0.77	0.44	A						
	500 MPs+ 10% LD50	0.77	0.49	1.56	S						
	1000 MPs+ 10% LD50	0.46	0.72	0.64	A						

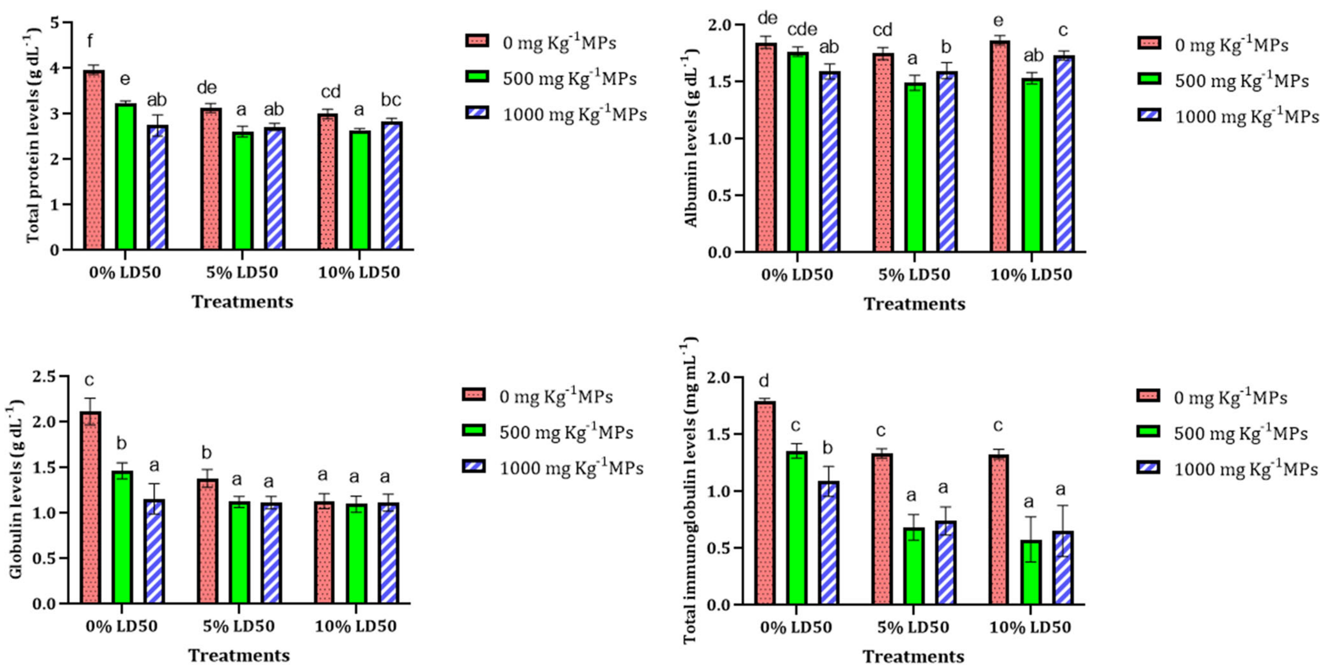
Predicated effect: PE; observed effect: OE; synergy ratio: SR; combined effect: CE; synergistic effect: S; suppressive effect: A

^{5} 5% LD_{50}: 2.5×10 CFU L^{-1} ; 10% LD_{50}: 5×10^{5} CFU L^{-1}



- 5% LD_{50} : 2.5×10^5 CFU L^{-1} ; 10% LD_{50} : 5×10^5 CFU L^{-1}

Fig. 2 Changes of cholesterol, triglyceride, high-density lipoprotein, and low-density lipoprotein levels in the blood of rainbow trout after exposure to MPs and *Y. ruckeri*. Results are illustrated as mean ± SD. Significant differences between groups were identified by alphabetical characters ($P < 0.05$). Different letters of the alphabet indicated the existence of significant differences between different groups



- 5% LD_{50} : 2.5×10^5 CFU L^{-1} ; 10% LD_{50} : 5×10^5 CFU L^{-1}

Fig. 3 Changes of total protein, albumin, globulin, and total immunoglobulin levels in the blood of rainbow trout after exposure to MPs and *Y. ruckeri*. Results are illustrated as mean ± SD. Significant differences between groups were identified by alphabetical characters ($P < 0.05$). Different letters of the alphabet indicated the existence of significant differences between different groups

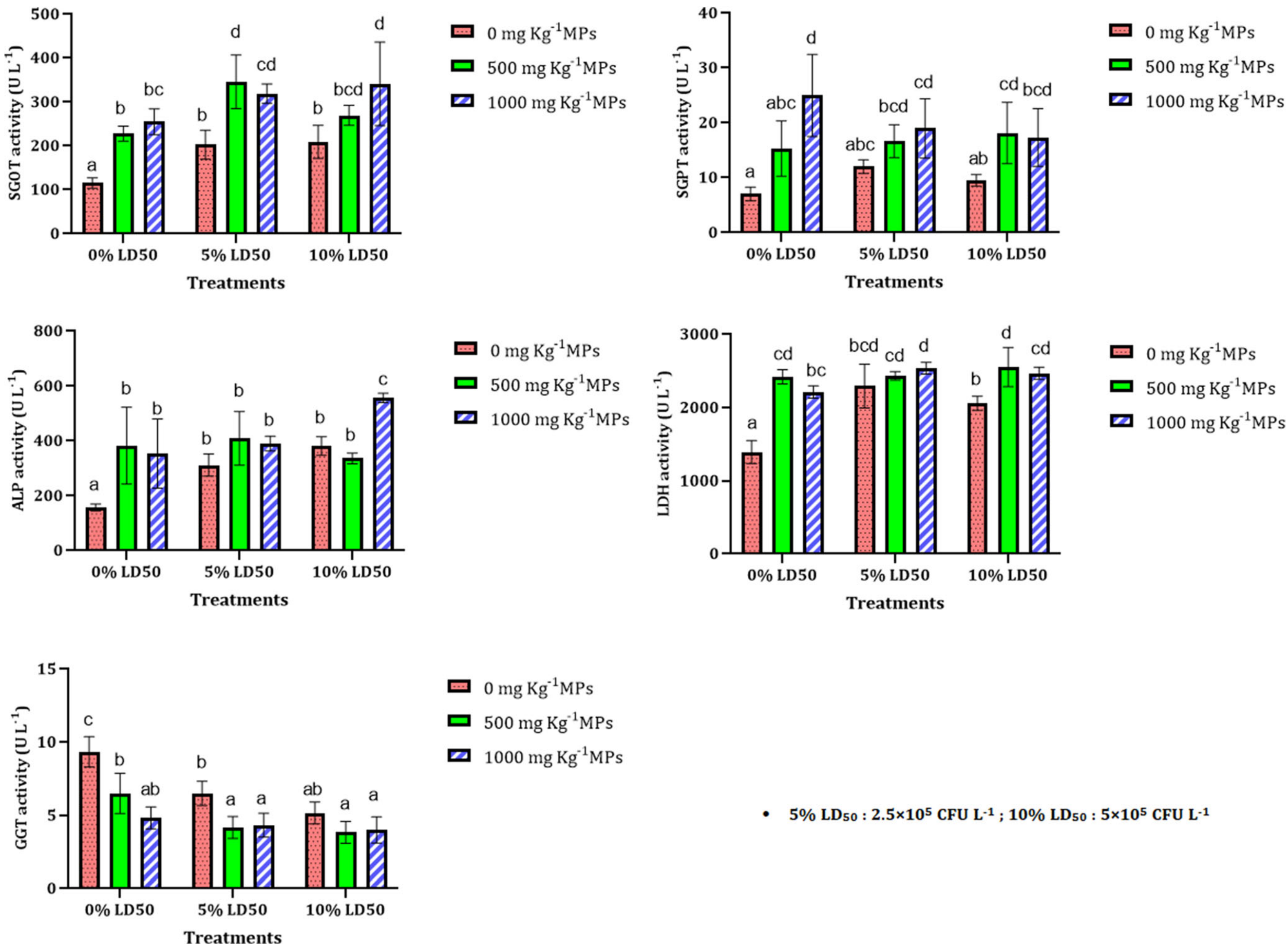


Fig. 4 Changes of SGOT, SGPT, ALP, LDH, and GGT activities in the blood of rainbow trout after exposure to MPs and *Y. ruckeri*. Results are illustrated as mean ± SD. Significant differences between groups were identified by alphabetical characters ($P < 0.05$). Different letters of the alphabet indicated the existence of significant differences between groups

fish showed an inverse trend with triglyceride concentration, which may be related to its different biological role. Cholesterol is not known as the primary energy source but is involved in the biosynthesis of biological membranes, steroid and corticosteroid hormones, and bile acids (Wen et al. 2021). Therefore, cholesterol depletion may be attributed to impaired cholesterol biosynthesis in the liver or its absorption in the intestine, increased biosynthesis rates of corticosteroid hormones, and bile acids. Porins and lipopolysaccharides play an essential role in increasing cytokinin biosynthesis in hepatocytes and macrophages (Portnyagina et al. 2021). Hence, the liver excretes cytokines by enhancing bile acids' biosynthesis to maintain biochemical homeostasis (Li et al. 2006). A significant decrease in cholesterol levels was observed in the serum of African catfish, *Clarias gariepinus* exposed to MPs (Kim et al. 2021).

High-density lipoprotein concentrations reduced significantly after fish exposure to *Y. ruckeri* and MPs alone and in combination. A significant decline was detected in low-density lipoprotein concentrations in the plasma of fish after co-exposure to MPs and *Y. ruckeri*. Decreased LDL and HDL may be due to lower blood cholesterol levels (Fig. 2 and Table 1).

Total protein, globulins, and total immunoglobulins content in the plasma decreased significantly following fish exposure to *Y. ruckeri* and MPs alone and in combination. The globulin and total immunoglobulin levels in fish exposed to both MPs and bacteria were considerably lower than the other experimental groups. Co-exposure to MPs and *Y. ruckeri* resulted in decreased albumin in fish (Fig. 3 and Table 1).

Decreased total protein may be due to increased protein catabolic rate to counteract the toxicity of MPs and bacterial biotoxins. Furthermore, MPs may inhibit the absorption of essential amino acids and reduce food digestibility in the aquatic digestive system. Decreases in globulin, albumin, and immunoglobulin levels also reflect a decrease in total plasma protein. Damage to the intestinal epithelium of fish after challenge with *Y. ruckeri* may also prevent the absorption of essential amino acids. Furthermore, Fernandez et al. (2003) found that Yrp1 protease plays a critical role in

Table 2 Evaluation of synergism of MPs and *Y. ruckeri* concerning the plasma enzyme activities and oxidative biomarkers

Parameters	Treatment	PE	OE	SR	CE	Parameters	Treatment	PE	OE	SR	CE
SGOT	500 MPs+ 5% LD50	3.49	3.01	1.16	S	CAT	500 MPs+ 5% LD50	0.89	0.64	1.40	S
	1000 MPs+ 5% LD50	3.91	2.78	1.41	S		1000 MPs+ 5% LD50	1.09	0.60	1.82	S
	500 MPs+ 10% LD50	3.61	2.35	1.54	S		500 MPs+ 10% LD50	0.93	0.44	2.13	S
	1000 MPs+ 10% LD50	4.04	2.97	1.36	S		1000 MPs+ 10% LD50	0.70	0.78	0.90	A
SGPT	500 MPs+ 5% LD50	3.75	2.38	1.57	S	GPx	500 MPs+ 5% LD50	0.68	0.66	1.04	S
	1000 MPs+ 5% LD50	6.11	2.71	2.25	S		1000 MPs+ 5% LD50	0.65	0.60	1.09	S
	500 MPs+ 10% LD50	2.97	2.60	1.15	S		500 MPs+ 10% LD50	0.63	0.59	1.08	S
	1000 MPs+ 10% LD50	4.85	2.48	1.96	S		1000 MPs+ 10% LD50	3.08	2.68	1.15	S
ALP	500 MPs+ 5% LD50	4.78	2.59	1.84	S	SOD	500 MPs+ 5% LD50	4.03	2.98	1.35	S
	1000 MPs+ 5% LD50	4.42	2.47	1.79	S		1000 MPs+ 5% LD50	3.99	3.05	1.31	S
	500 MPs+ 10% LD50	5.84	2.13	2.75	S		500 MPs+ 10% LD50	5.22	3.28	1.59	S
	1000 MPs+ 10% LD50	5.40	3.53	1.53	S		1000 MPs+ 10% LD50	0.59	0.86	0.69	A
LDH	500 MPs+ 5% LD50	2.87	1.75	1.64	S	MDA	500 MPs+ 5% LD50	0.60	0.73	0.82	A
	1000 MPs+ 5% LD50	2.63	1.83	1.44	S		1000 MPs+ 5% LD50	0.53	0.69	0.76	A
	500 MPs+ 10% LD50	2.58	1.84	1.40	S		500 MPs+ 10% LD50	0.53	0.56	0.95	A
	1000 MPs+ 10% LD50	2.36	1.78	1.33	S		1000 MPs+ 10% LD50	170	35.00	4.88	A
GGT	500 MPs+ 5% LD50	0.49	0.45	1.09	S	Total antioxidant	500 MPs+ 5% LD50	212	36.52	5.82	S
	1000 MPs+ 5% LD50	0.36	0.46	0.78	A		1000 MPs+ 5% LD50	197	42.67	4.62	S
	500 MPs+ 10% LD50	0.38	0.41	0.93	A		500 MPs+ 10% LD50	245	47.24	5.19	S
	1000 MPs+ 10% LD50	0.28	0.43	0.66	A		1000 MPs+ 10% LD50	0.89	0.64	1.40	S

Predicated effect: PE; observed effect: OE; synergy ratio: SR; combined effect: CE; synergistic effect: S; suppressive effect: A

^{5} 5% LD_{50}: 2.5×10 CFU L^{-1} ; 10% LD_{50}: 5×10^5 CFU L^{-1}

biodegradation of a wide variety of matrix and muscle proteins (Fernandez et al. 2003). Therefore, the increased biodegradation rate of plasma proteins could decrease total proteins, albumin, and globulin. Total proteins, albumin, and globulin levels decreased in the serum of freshwater pond turtles, *E. orbicularis* exposed to MPs (Banaee et al. 2021).

AST (SGOT) and ALT (SGPT) are non-specific enzymes found in two different cytoplasmic and mitochondrial isoforms in all tissues (McGill 2016; Wang and Chen 2018). SGOT and SGPT play an essential role in glutathione biosynthesis and glycogen renewal in liver cells (Ellinger et al. 2011). Furthermore, aminotransferases maintain a balance between NAD^+/NADH in cells (McGill 2016). The role of aminotransferases in glycerol regeneration in adipose tissues and neurotransmitter biosynthesis in the neuroglia is considerable (Wang and Chen 2018). Exposure of fish to MPs and *Y. ruckeri*, alone and in combination, resulted in a significant increase in SGOT, ALP, and LDH activities. Challenge of fish with *Y. ruckeri* had no significant effect on SGPT activity (Fig. 4 and Table 2). Damage to hepatocytes may lead to increased SGOT, SGPT, ALP, and LDH activities. The results showed that SGPT activity significantly increased in fish exposed to 1000 mg Kg^{-1} MPs (Fig. 4 and Table 2). Furthermore, combined exposure to MPs and bacteria increased SGPT activity in fish plasma. The change in SGOT, SGPT, LDH, and ALP activities can be attributed to the hepatotoxicity effect of bacterial lipopolysaccharide (Beheshti et al. 2021). Also, porins can significantly affect cellular biochemical homeostasis by disrupting the selective permeability of cell membranes (Portnyagina et al. 2021). ALP is a transmembrane metalloenzyme involved in protein phosphorylation, cell growth, apoptosis, and cell migration (Derikvandy et al. 2020). Therefore, any damage to the cell membrane can lead to a change in ALP activity in the blood (Banaee et al. 2019b). ALP activity significantly increased in the serum of *Symphysodon aequifasciatus* and *O. niloticus* after exposure to MPs (Wen et al. 2018; Hamed et al. 2019). SGOT, SGPT, and ALP activity increased in turtles' serum (*Emys orbicularis*) after MPs' exposure (Banaee et al. 2021).

LDH plays a vital role in converting lactate to pyruvate, NAD^+ to NADH, and vice versa (Soleimany et al. 2016). LDH is an indicator of hypoxia and mitochondrial oxidation function. Also, an increase in extracellular LDH can indicate cell death or necrosis (Maes et al. 2015). Increased LDH activity was detected in the *C. carpio* and *S. aequifasciatus* exposed to MPs combined with Cd (Wen et al. 2018; Banaee et al. 2019a). GGT is an anchorage enzyme in the cell membrane that plays a unique role in the gamma-glutamyl cycle in the biosynthesis and biodegradation of glutathione and detoxification from xenobiotics (Hatami et al. 2019). Compared

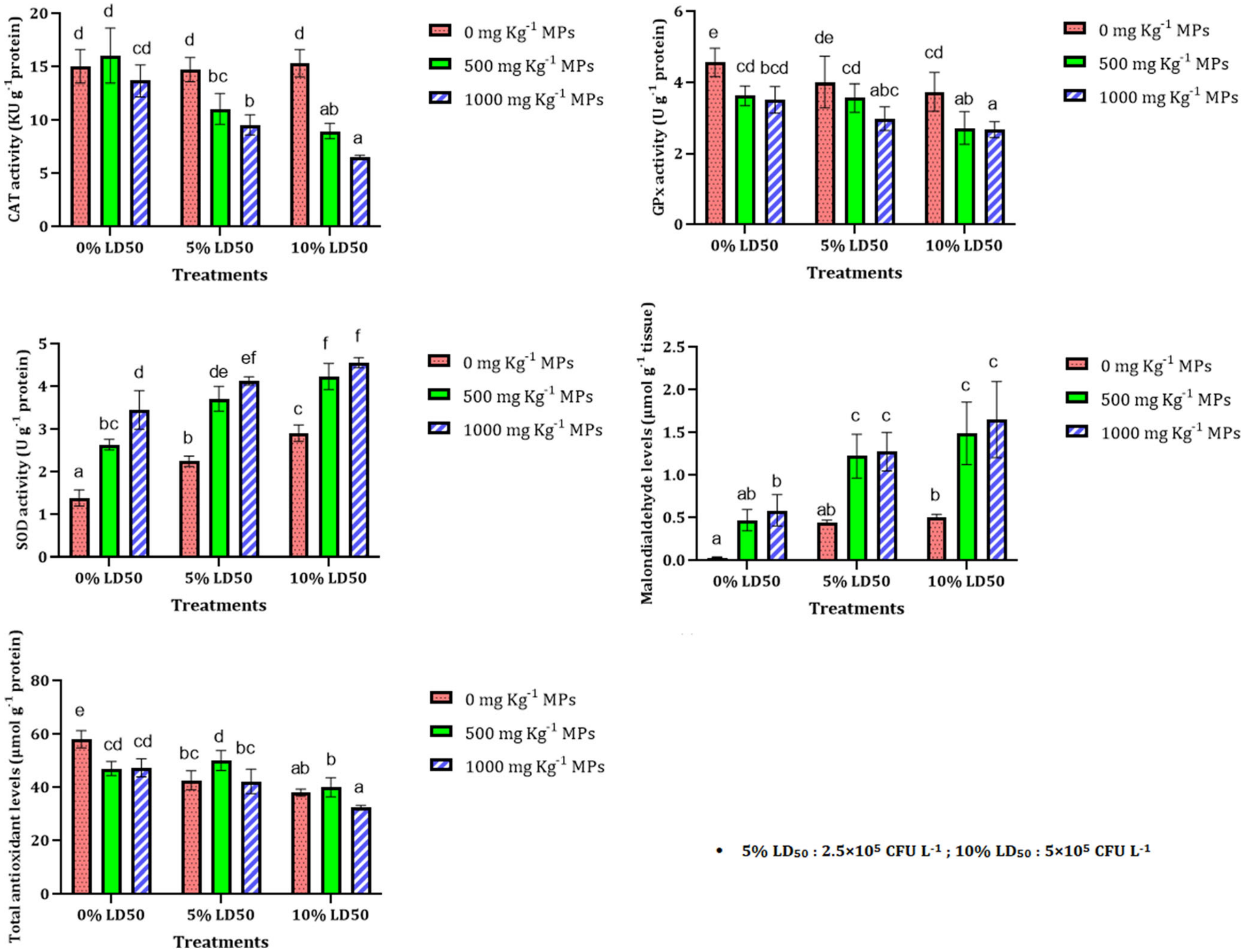


Fig. 5 Changes of CAT, GPx, SOD activities, and MDA and total antioxidant levels in the hepatocyte of rainbow trout after exposure to MPs and *Y. ruckeri*. Results are illustrated as mean ± SD. Significant differences between groups were identified by alphabetical characters ($P < 0.05$). Different letters of the alphabet indicated the existence of significant differences between groups

with the reference group, GGT activity decreased in the plasma of fish exposed to MPs and *Y. ruckeri* individually and concurrently (Fig. 4 and Table 2). Changes in the catalytic domain of the GGT can lead to its inactivation. GGT is often degraded after inactivation, and its metabolites are excreted in the bile (Fornaciari et al. 2014). The relationship between decreased GGT activity in the blood and increased biliary excretion of its metabolites has also been demonstrated (Li and Chiang 2014). Decreased GGT activity in serum of freshwater pond turtles (*E. orbicularis*) exposed to MPs may indicate a reduction in detoxification capacity in hepatocytes (Banaee et al. 2021). Damage to the intestinal epithelium after bacterial challenge and exposure to MPs may lead to gastrointestinal disorders. Therefore, a reduction in GGT activity can be attributed to the disturbance in the absorption of minerals and vitamins.

A significant decrease was observed in CAT activity in the hepatocyte of fish after co-exposure to MPs and *Y. ruckeri*. The result shows that MPs combined with *Y. ruckeri* inhibited CAT (Fig. 5 and Table 2). CAT plays a vital role in the breakdown of hydrogen peroxide (Banaee et al. 2013). Therefore, inhibition of CAT activity may lead to an increase in H_2O_2 at the cellular level. However, there was no significant difference in CAT activity between the MPs, bacterial groups alone and the control group. A decrease in the CAT activity was reported in the *Oryzias melastigma* and *Danio rerio* following exposure to the MPs (Wang et al. 2019; Wan et al. 2019). GPx activity in hepatocytes of fish challenged with 10% LD_{50} *Y. ruckeri* was significantly lower than the control group. Exposure of fish to MPs alone and MPs combined with *Y. ruckeri* reduced GPx activity in hepatocytes (Fig. 4 and Table 2). GPx is an enzyme involved in GSH-related antioxidant defense mechanisms that can neutralize organic peroxides (ROOH) and hydroperoxides (Banaee et al. 2013). Therefore, decreased GPx activity can lead to oxidative stress in fish exposed to MPs and *Y. ruckeri*. The GPx activity significantly decreased in *D. rerio* after exposure to MPs (Umamaheswari et al. 2020). Significant activation of SOD occurred after fish exposure to MPs and *Y. ruckeri*, alone and in combination (Fig. 5 and Table 2). SOD is the only antioxidant enzyme that neutralizes the superoxide anion by transforming this ROS to oxygen and hydrogen peroxide (Banaee et al. 2013). Increased SOD activity may be a physiological response to enhanced superoxide anions in cells. SOD activity significantly increased in the liver of goldfish, *Carassius auratus*, and common carp, *C. carpio*, after exposure to MPs (Xia et al. 2020; Yang et al. 2020).

Exposure to MPs and *Y. ruckeri,* alone and in combination, resulted in a significant decrease in total antioxidant levels in fish's hepatocytes. MDA content in the hepatocytes of fish exposed to 500 mg Kg^{-1} MPs and 5% LD_{50} *Y. ruckeri* alone did not show any significant changes to the control group. However, MDA content was increased considerably in the hepatocytes of fish exposed to 1000 mg Kg^{-1} MPs and 10% LD_{50} *Y. ruckeri* alone. Co-exposure of fish to MPs and *Y. ruckeri* increased MDA levels. Raised lipid peroxidation levels indicate an increase in the rate of reactive oxygen species production (Fig. 4 and Table 2). A significant increase in MDA was observed in the *O. niloticus* (Hamed et al. 2019), (Ding et al. 2018), *S. aequifasciatus* (Wen et al. 2018), *C. auratus* (Xia et al. 2020), *C. carpio* (Yang et al. 2020), *D. rerio* (Umamaheswari et al. 2020), and *Dicentrarchus labrax* (Barboza and Guilhermino 2018).

Changes in oxidative stress biomarkers indicated oxidative stress in fish hepatocytes in the challenge with *Y. ruckeri* and MPs. Exposure of fish to MPs can disrupt the balance between reactive oxygen species' production rate and cells' antioxidant defense capacity. Hence, when the antioxidant defense capacity is insufficient to quench the ROS, it will lead to oxidative damage (Fig. 5 and Table 2).

Ryckaert et al. (2010) and Guijarro et al. (2018) found that in vitro inoculation of rainbow trout renal macrophages with *Y. ruckeri* can induce reactive oxygen species production. Thus, changes in antioxidant enzyme activities and cellular antioxidant capacity may have been due to ROS increase by macrophages of fish after challenge with *Y. ruckeri*. Moreover, bacterial metalloproteinases may activate mechanisms involved in producing reactive oxygen species that could lead to oxidative stress (Dasgupta et al. 2010).

Conclusion

In conclusion, exposure of fish to MPs resulted in changes in blood biochemical parameters and biomarkers of oxidative stress. Changes in the biochemical parameters and oxidative biomarkers may be due to cytotoxic effects of MPs. Disruption of biochemical homeostasis in fish challenged with *Y. ruckeri* may be related to the potential for bacterial pathogenesis. Therefore, this study's biochemical indicators can be considered clinical signs of fish treated with *Y. ruckeri*. This study showed that exposure to MPs and *Y. ruckeri* can significantly change biochemical indices and cause oxidative stress. This study revealed the synergistic effect of MPs in exacerbating the clinical symptoms of fish in the challenge with *Y. ruckeri*. These results suggest that the cytotoxic effects of xenobiotics such as MPs may increase the pathogenicity of *Y. ruckeri* when fish are exposed to both components.

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Author contribution *Banihashemi: PhD. Student*; contribution: investigation, project administration

Amin Gholamhosseini; Assistant Professor (Veterinary); contribution: super-adviser; cooperation in project implementation, writing—original draft

Siyavash Soltanian; Professor (Veterinary); contribution: adviser, cooperation in project implementation

Mahdi Banaee; Assistant Professor (Aquaculture and Ecotoxicology); contribution: adviser, validation, formal analysis, writing—original draft

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Data availability The data that support the findings of this study are available at Shiraz University but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Shiraz University.

Declarations

Ethics approval and consent to participate In this study, none of the authors used human beings as research subjects. International, national, and institutional guidelines for the care and use of animals were followed. Experimental protocols were done following the Iranian Animal Ethics Framework under the supervision of the Iranian Society for the Prevention of Cruelty to Animals and Shiraz University Research Council (SURC973/46-2679-2).

Consent to publish In the present study, no personal information is used (including any individual's personal details, images or videos).

Conflict of interest The authors declare that they have no conflict of interest.

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